

CONSOLIDATED ENGINEERING AND SOURCING CONSULTANTS (CESC)

Engineering White Paper

Busbar Protection Engineering

Low-Impedance and High-Impedance Schemes, CT Performance, Stability, and Practical Selection

Prepared, reviewed, and issued by CESC

Website publication edition

Executive abstract

This paper compares low-impedance and high-impedance busbar protection schemes, with emphasis on CT performance, stability during external faults, scheme selection, and engineering checks relevant to grid-station design.

Document basis

Item	Description
Document type	Technical white paper
Primary basis	IEC 60255-187-1, IEC 61869-2, IEC 60909, IEC 62271-200
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	CESC engineering capability in studies, protection, digital integration, retrofit, and asset management.

Applicable references

Reference	Application in this paper
IEC 60255-187-1	Differential protection functional requirements context.
IEC 61869-2	Current-transformer application basis.
IEC 60909	Short-circuit duty and study basis context.
IEC 62271-200	MV switchgear context and station implementation environment.

Purpose

Busbar faults require fast and secure clearance because they affect multiple circuits and can escalate mechanical and thermal damage quickly. Busbar protection engineering therefore demands a higher level of CT assessment and zone definition than ordinary feeder protection.

The design choice between low-impedance and high-impedance methods should reflect bus arrangement, CT quality, expansion requirements, and maintenance philosophy.

Scheme comparison

High-impedance schemes remain attractive for simplicity and proven security where CT matching and circuit arrangement are suitable. Low-impedance numerical schemes provide greater adaptability for complex multi-zone busbars, breaker-and-a-half arrangements, and future extension.

The engineer must assess not only relay functionality but also CT saturation exposure, wiring complexity, and commissioning maintainability.

Key calculations

For high-impedance applications, the stabilizing voltage and CT knee-point checks remain central. The CT knee-point requirement is commonly assessed against the total loop resistance of CT secondary winding, leads, and relay or stabilizing resistor path.

For any busbar design, the maximum external fault through-current must be reviewed because this is often the most severe CT saturation duty even though the fault is outside the protected zone.

Illustrative engineering calculations

The following simplified checks are typical of the engineering work carried out during busbar protection design. The figures are illustrative and should be replaced by project-specific study values during detailed design.

Engineering check	Illustrative calculation / interpretation
1) External fault through-current review	Assume a 33 kV bus section with calculated maximum three-phase external fault current of 31.5 kA rms for 1 s. For CT adequacy review, this external through-current is often more onerous than an internal bus fault because the scheme must remain stable while CTs may be driven toward saturation. Primary fault level check: $S = 1.732 \times V \times I = 1.732 \times 33 \text{ kV} \times 31.5 \text{ kA} = \text{approximately } 1,800 \text{ MVA}$ short-circuit level at the bus. This confirms that busbar CT performance and relay stability must be checked against a severe network duty rather than nominal load current.
2) High-impedance stabilizing voltage check	For a high-impedance scheme, a common first-pass check is $V_{stab} \geq I_{f_ext} \times (R_{ct} + 2R_l + R_r)$, where I_{f_ext} is the maximum external fault secondary current referred to the CT secondary, R_{ct} is CT secondary resistance, R_l is one-way pilot lead resistance, and R_r is relay path resistance. Example: CT ratio 2000/1 A, external fault current 31.5 kA primary, CT secondary current $I_{f_ext} = 31.5/2000 \times$

	1000 = 15.75 A secondary. Assume $R_{ct} = 6 \Omega$, lead loop resistance $2R_l = 1.5 \Omega$, and relay path resistance $R_r = 0.5 \Omega$. Then $V_{stab} \geq 15.75 \times (6 + 1.5 + 0.5) = 15.75 \times 8 = 126 \text{ V}$. In practice, the selected stabilizing setting is usually increased above the calculated minimum to provide security margin for CT errors, wiring tolerance, and transient DC offset.
3) CT knee-point check for high-impedance protection	A typical engineering rule is that CT knee-point voltage V_k should comfortably exceed approximately $2 \times V_{stab}$ for stable performance under offset external faults, with the exact requirement verified against the selected relay and application standard. Using the above example, minimum practical $V_k \geq 2 \times 126 = 252 \text{ V}$. The engineer would therefore avoid marginal CTs and typically select a higher knee-point class, for example 400 V or above, subject to the detailed scheme requirement and manufacturer data.
4) Stabilizing resistor sizing concept	If the relay requires a higher effective operating voltage than the natural loop resistance provides, an external resistor is added so that $R_{stab} = (V_{set} / I_{set}) - R_{loop}$, where R_{loop} is the total existing secondary circuit resistance. Example: if a design target of 150 V at 1 A relay operating current is adopted and the measured loop resistance is 8Ω, then $R_{stab} = 150/1 - 8 = 142 \Omega$. The resistor thermal rating must also be checked for the worst credible spill current duration.
5) Low-impedance differential setting concept	For numerical low-impedance busbar differential protection, a basic design check is to ensure that minimum internal fault spill current exceeds the chosen differential pickup while maximum external fault spill remains below the slope characteristic. Illustrative pickup check: if minimum internal fault spill current at the relay is estimated at 0.45 A secondary, a conservative pickup may be set at 0.20 A to 0.25 A, leaving adequate sensitivity margin while still allowing stability logic and bias slope to restrain during heavy through-faults. Illustrative slope concept: for an external through-current producing 16 A restraint current, a 30 percent biased element would permit up to 4.8 A differential before operation. This helps explain why low-impedance numerical schemes can tolerate moderate CT mismatch while remaining secure.

6) Busbar short-time thermal withstand awareness	Protection engineering should remain aligned with switchgear short-time duty. A 31.5 kA, 1 s busbar has thermal energy exposure proportional to $I^2t = 31.5^2 \times 1 = 992.25 \text{ kA}^2\text{s}$. Fast busbar trip times reduce the actual thermal and electrodynamic stress imposed on conductors, supports, and adjacent compartments.
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Engineering interpretation

These calculations do not replace manufacturer-specific relay manuals or detailed CT excitation studies, but they materially strengthen front-end scheme selection. Including such checks in design submissions demonstrates that the busbar protection philosophy has been developed from fault duty, CT behaviour, and station configuration rather than from relay catalogue features alone.

Practical design notes

Zone selection should follow the physical bus arrangement, isolator logic, and maintenance states. Incorrect treatment of isolator position and zone transfer logic is one of the most common causes of avoidable design risk in busbar schemes.

The design file should include zone diagrams, CT allocation tables, bay status logic, blocking philosophy, trip matrix, and maintenance inhibit procedures.

Execution and extension

Busbar schemes should be engineered for future bays, not only the day-one arrangement. Spare CT core philosophy, panel I/O headroom, and zone numbering discipline all matter if the station is likely to be extended later.

Testing should include external-fault stability, internal-zone operation, isolator transfer states, and restoration after bay outage or panel substitution.

CESC engineering contribution

CESC can support busbar protection selection, CT adequacy review, logic-zone preparation, trip matrix definition, and brownfield extension planning with an emphasis on both protection security and constructability.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. The intended message to clients is that CESC can define studies, design criteria, scheme architecture, implementation strategy, testing approach, and brownfield execution method across protection, control, digital integration, retrofit, and asset management scopes.